

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (AIML/CCE) (Credit System) Degree Examinations**

May - June 2023

20AM623 / 20CC623 - AUGMENTED AND VIRTUAL REALITY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	List modern VR experiences. Explain any six of them.	8	L*2	1	1
b)	Write short note on the following with respect to virtual reality. i) First vs. third person ii) Synthetic vs. captured world iii) Health and safety	6	L2	1	1
c)	Explain the three conditions to be satisfied by the rotational matrix for 2-D rotation. Derive the relation between 2-D rotational matrix and circle with unit radius centered at the origin.	6	L3	1	2
2. a)	Explain how the idea of 2-dimensional rotation is extended for 3-dimensional rotation by obtaining matrices for roll, pitch and yaw.	6	L3	2	2
b)	Write a short note on chaining of transformations. Discuss transformations applied for left and right eyes.	5	L3	2	2
c)	List and explain the components of look-at. Show the relation between Eye transformation T_{eye} and these components.	5	L2	2	1
d)	Explain canonical view transform.	4	L2	2	1
3. a)	Explain the following terminologies. i) Foveated rendering, ii) Display scanout, iii) Vergence accommodation mismatch with examples.	8	L2	2	1
b)	Define quaternion. Show the relation between quaternion and axis-angle rotation. Convert the following to axis angle representations / quaternion $(0, 0, 0, 1)$, $(1/\sqrt{2}, 0, 1/\sqrt{2}, 0)$, $((1, 0, 0), \pi)$, $((1, 0, 0), \pi/2)$.	6	L3	2	2
c)	Explain different type of eye movements with examples.	6	L2	2	1
Unit – II					
4. a)	Discuss different shading methods used for visual rendering with respect to image order rendering.	8	L2	3	1
b)	Discuss the following with respect to rasterization. i) Aliasing ii) Culling	6	L2	3	1
c)	Explain the following in connection with visual rendering. i) Simplifying the virtual world ii) Post-rendering image warp	6	L2	3	1
5. a)	List and explain the four general problems that must be solved to make an effective tracking system with respect to 2D orientation.	6	L2	3	1
b)	List and explain different types of vection.	6	L2	3	1
c)	Explain the Detection mechanism with respect to perception of motion.	4	L2	3	1
d)	Explain the Painter's algorithm and its limitation. How can we overcome the limitation with the help of depth buffer?	4	L2	3	1

P.T.O.

6. a) Write a note on the following with respect to tracking 3D orientation.

i) Integration

ii) Tilt correction and Yaw correction

- b) Write a short note on tracking 2D orientation.

- c) List and explain the factors that affect sensitivity.

8	L2	4	1
6	L2	4	1
6	L2	4	1

Unit – III

7. a) Discuss different marker types used in augmented reality applications.

- b) List and explain the applications of augmented reality.

- c) Define augmented reality. With block diagram explain the components required for AR.

- d) List all the steps involved in marker detection procedure.

7	L2	5	1
6	L2	5	1
4	L2	5	1
3	L2	5	1

8. a) Briefly discuss the scenarios where marker based solution is a good choice.

- b) Briefly explain each of the following:

i) Model based tracking

ii) Sensor tracking

iii) Hybrid tracking

- c) Explain the types of IR markers. Write down the applications and limitations of infrared markers.

8	L2	5	1
6	L2	5	1
6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
